

## Tic-Tac-Toe Report

### Trace of moves:

```
|||  
|||  
|||
```

Turn: X

Player X took position (0, 2).

```
|||X|  
|||  
|||
```

Turn: O

reference:

row 0 is neutral.

row 1 is happy.

row 2 is surprise.

Emotion detected as surprise (row 2). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

reference:

col 0 is neutral.

col 1 is happy.

col 2 is surprise.

Emotion detected as neutral (col 0). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

Player O took position (2, 0).

```
|||X|  
|||  
|O||
```

Turn: X

Player X took position (2, 2).

```
|||X|  
|||  
|O|X|
```

Turn: O

reference:

row 0 is neutral.

row 1 is happy.

row 2 is surprise.

Emotion detected as neutral (row 0). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

reference:

col 0 is neutral.

col 1 is happy.

col 2 is surprise.

Emotion detected as neutral (col 0). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

Player O took position (0, 0).

```
|O| |X|
```

```
||| |
```

```
|O| |X|
```

Turn: X

Player X took position (2, 1).

```
|O| |X|
```

```
||| |
```

```
|O|X|X|
```

Turn: O

reference:

row 0 is neutral.

row 1 is happy.

row 2 is surprise.

Emotion detected as neutral (row 0). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

text

1

reference:

col 0 is neutral.

col 1 is happy.

col 2 is surprise.

Emotion detected as neutral (col 0). Enter 'text' to use text input instead (0, 1 or 2). Otherwise, press Enter to continue.

Player O took position (1, 0).

```
|O| |X|
```

```
|O| | |
```

```
|O|X|X|
```

Player O has won!

### **How well did your interface work?**

For the most part our interface worked well and was able to recognize surprised and neutral emotions very well. However, we ran into issues with having any happy emotions to be successfully recognized by the model during the tic-tac-toe game. For the technical side of the interface, the camera UI worked very well and there was a fast response time from capture to facial recognition output

### **Did it recognize your facial expressions with the same accuracy as it achieved against the test set?**

For the emotions of neutral and surprise, we think that it had similar or higher accuracy as the test set since our model was able to reliably recognize those two emotions during the tic-tac-toe game. However, the accuracy for the happy emotion was a lot worse than against the test set. One of the potential reasons that the model wasn't able to recognize happy emotions during the tic-tac-toe game could be that any objects in the background of the webcam could have affected the recognition of the emotion.

### **Lines of Code Added for `_get_emotion` function:**

```
def _get_emotion(self, img) -> int:
    import cv2
    import numpy as np
    from tensorflow.keras.models import load_model

    if not hasattr(self, '_model'):
        self._model =
load_model('results/basic_model_30_epochs_timestamp_1785804752.keras')

    resized = cv2.resize(img, image_size)
    rgb = cv2.cvtColor(resized, cv2.COLOR_GRAY2RGB)
    batch = np.expand_dims(rgb, axis=0)

    prediction = self._model.predict(batch, verbose=0)
    emotion = int(np.argmax(prediction[0]))

    return emotion
```